Wireless IoT and LAN (CESE4060): Matlab or NS3 Report

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March 25, 2026

Report contains 4 of maximum 4 pages.

1 Basic Idea

This work evaluates **IEEE 802.11ac** (Wi-Fi 5, TDMA) and **IEEE 802.11ax** (Wi-Fi 6, OFDMA) through two system-level scenarios simulated in MATLAB. The core architectural difference is the multiple-access scheme: 802.11ac grants the full 80 MHz channel to one user at a time in round-robin, while 802.11ax divides it into four simultaneous 20 MHz Resource Units (RUs), one per user [1].

The intuition is that TDMA excels when all users share similar channel conditions and large payloads, as no bandwidth is lost to sub-band splitting. OFDMA excels when users differ in payload size or SNR, since each is served independently and head-of-line blocking is eliminated [2]. **Scenario A** stresses the delay behaviour with mixed large/small payloads; **Scenario B** stresses capacity and fairness with asymmetric SNR (Near vs. Far users).

2 Implementation

All scripts use MATLAB's WLAN Toolbox with LDPC coding, TGax Model-B fading channel, CBW80 (80 MHz), and 50 packets per user. Four independent `wlanTGaxChannel` objects are created at distances {3, 5, 7, 9} m.

Traffic profiles. Scenario A: two Video users (1500 B, $\mathrm{MCS_{ac}}$=8, $\mathrm{MCS_{ax}}$=10, SNR=30 dB) and two Voice users (128 B, $\mathrm{MCS_{ac}}$=4, $\mathrm{MCS_{ax}}$=5, SNR=25 dB). Scenario B: two Near users (1024 B, $\mathrm{MCS_{ac}}$=9, $\mathrm{MCS_{ax}}$=11, SNR=38/34 dB) and two Far users (512 B, $\mathrm{MCS_{ac}}$=2–1, $\mathrm{MCS_{ax}}$=3–2, SNR=12/8 dB).

802.11ac (TDMA). Each user is simulated with `wlanVHTConfig` (CBW80). The slot duration is measured from the waveform length divided by the sample rate. In round-robin, per-user throughput is $\eta_u = G_u/(T_u \times N)$, where $N = 4$ users share the channel time. The packet delay for user u (position u in the round) is modelled as: $queuing = (u-1) \times T_u$ (uses the user's own slot duration as a proxy for all preceding slots), plus its own transmission time T_u, plus propagation delay d_u/c.

802.11ax (HE-SU workaround). `wlanHEMUConfig` cannot be used for SISO chains because `NumSpaceTimeStreams` is read-only in multi-RU configurations and always defaults to a value exceeding the number of transmit antennas, causing a fatal error. The workaround simulates each user independently with `wlanHESUConfig` (CBW80, SISO). The OFDMA simultaneity is captured through time accounting: the shared PPDU slot equals the longest of all users' HE-SU frame durations (`ofdma_slot = max(slot_dur_ax)`). All users are charged this same clock duration per packet round, which is physically correct since in 802.11ax DL-OFDMA the AP transmits all RUs simultaneously and the frame length is set by the slowest RU. For 802.11ax, packet delay is $ofdma_slot + d_u/c$ (no queuing).

Spectral efficiency. Network-level SE divides total throughput by 80 MHz for both standards. Per-user SE for 802.11ax is referenced to the 20 MHz sub-band (`BW_HZ/N_USERS`); for 802.11ac it uses the full 80 MHz.

Metrics. Total and per-user throughput (Mbps); average packet delay (μs); network SE (bps/Hz); Jain's Fairness Index [3]: $J = (\sum x_i)^2/(N \sum x_i^2)$; normalised per-user throughput share.

3 Results

Scenario A — Heterogeneous Traffic (Small vs. Large Packets)

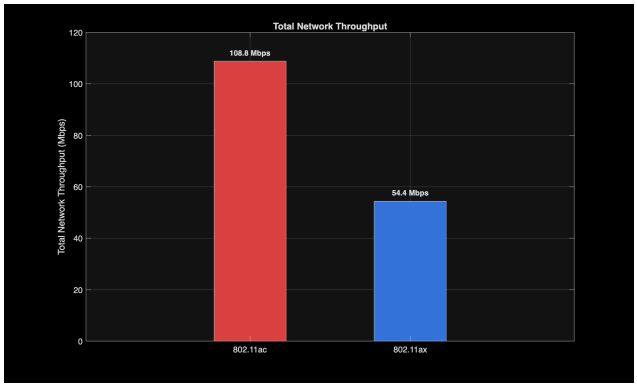

Figure 1: Scenario A: Total network throughput. 802.11ac (108.8 Mbps) achieves roughly double 802.11ax (54.4 Mbps). Splitting the 80 MHz band into four 20 MHz RUs with guard bands halves total capacity when all users have high SNR.

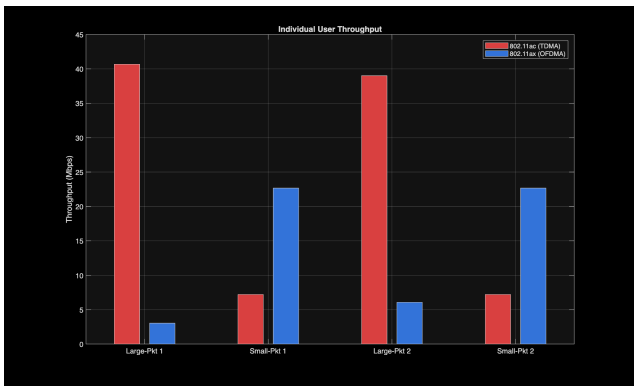

Figure 2: Scenario A: Per-user throughput. In 802.11ac, Large-Pkt users dominate (≈40 Mbps) while Small-Pkt users receive only ≈7 Mbps. In 802.11ax the pattern reverses: Small-Pkt users reach ≈22.6 Mbps in their dedicated 20 MHz RU, while Large-Pkt users fall to 3–6 Mbps.

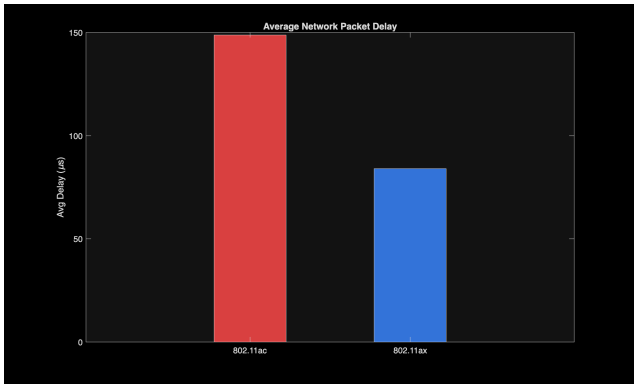

Figure 3: Scenario A: Average network packet delay. 802.11ax (≈83 μs) reduces network delay versus 802.11ac (≈148 μs) by eliminating queuing through parallel OFDMA scheduling.

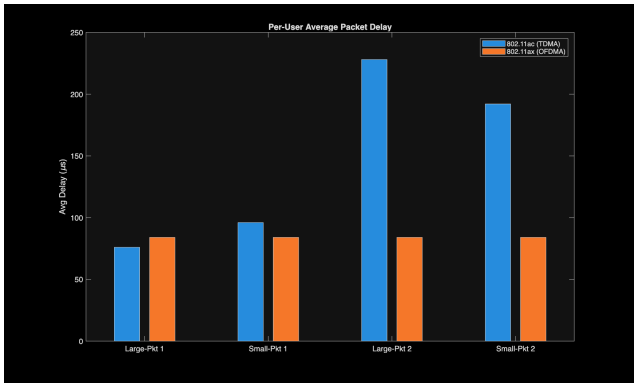

Figure 4: Scenario A: Per-user delay. 802.11ax enforces a uniform ≈84 μs for every user. In 802.11ac, Large-Pkt 2 (position 3 in round-robin) reaches 227 μs and Small-Pkt 2 (position 4) reaches 193 μs due to queuing.

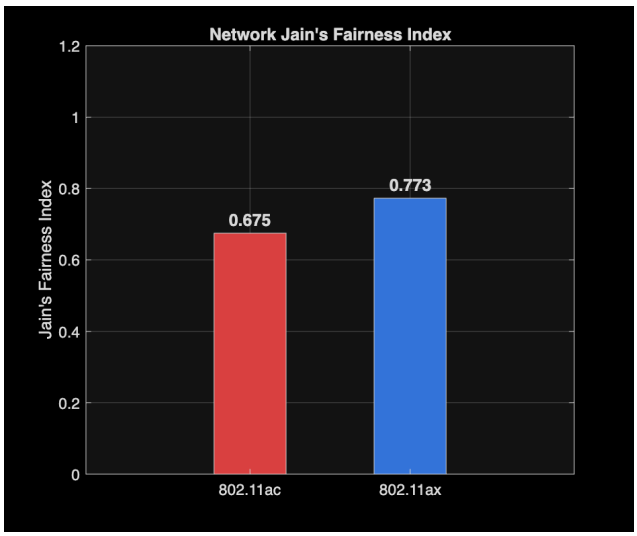

Figure 5: Scenario A: Jain's Fairness Index. 802.11ax (0.773) scores higher than 802.11ac (0.675), because OFDMA distributes throughput more evenly across traffic types.

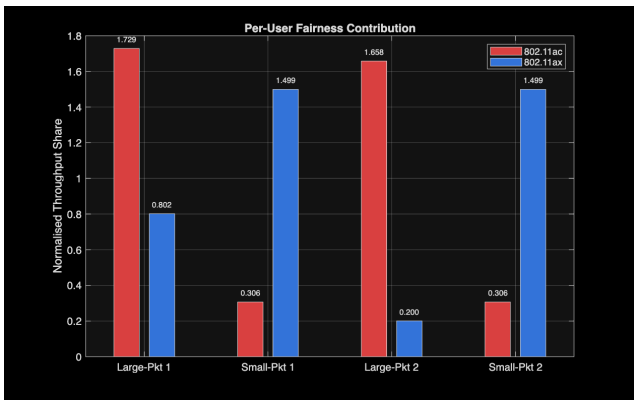

Figure 6: Scenario A: Per-user normalised throughput share. In 802.11ac, Large-Pkt users hold shares >1.6, while Small-Pkt users remain below 0.31. In 802.11ax, Small-Pkt users each reach 1.50, narrowing the disparity.

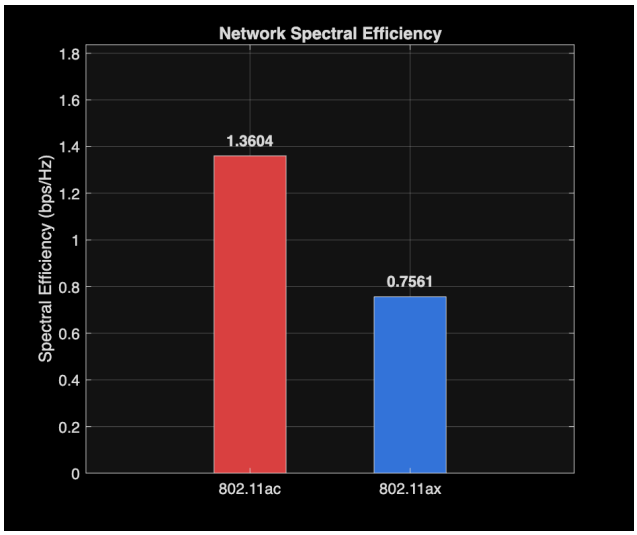

Figure 7: Scenario A: Network SE (referenced to 80 MHz). 802.11ac achieves 1.360 bps/Hz vs. 0.756 bps/Hz for 802.11ax, confirming the throughput penalty from RU sub-band splitting.

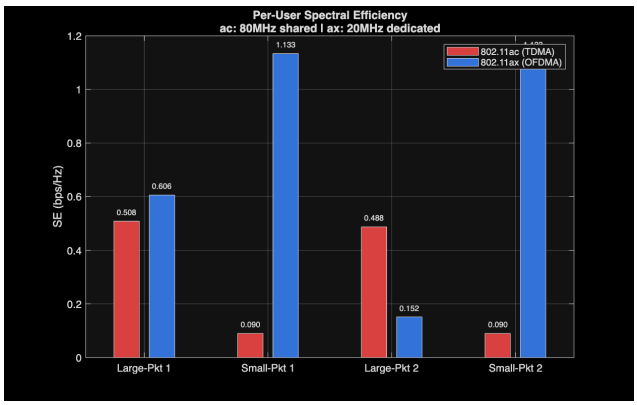

Figure 8: Scenario A: Per-user SE. For 802.11ax, SE is referenced to the 20 MHz sub-band per user; for 802.11ac to the full 80 MHz time-shared. Small-Pkt users in 802.11ax reach >1.1 bps/Hz per sub-band; Large-Pkt users in 802.11ac lead at ≈0.50 bps/Hz.

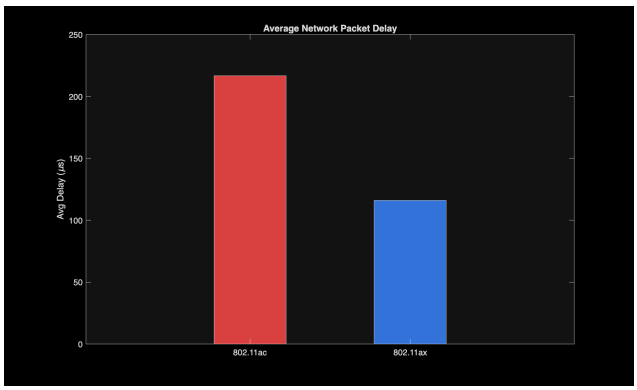

Figure 11: Scenario B: Average network delay. 802.11ax (≈115 μs) is significantly lower than 802.11ac (≈216 μs), driven by OFDMA's elimination of per-user queuing.

Scenario B — Variable SNR (Near vs. Far Users)

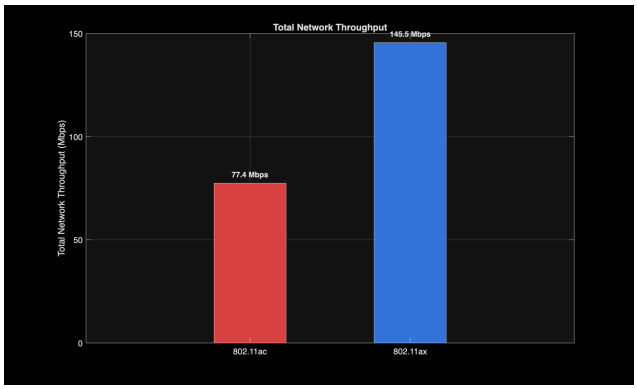

Figure 9: Scenario B: Total network throughput. 802.11ax (145.5 Mbps) is nearly double 802.11ac (77.4 Mbps): parallel OFDMA allows Near users to transmit at peak rates unimpeded by Far users in the TDMA queue.

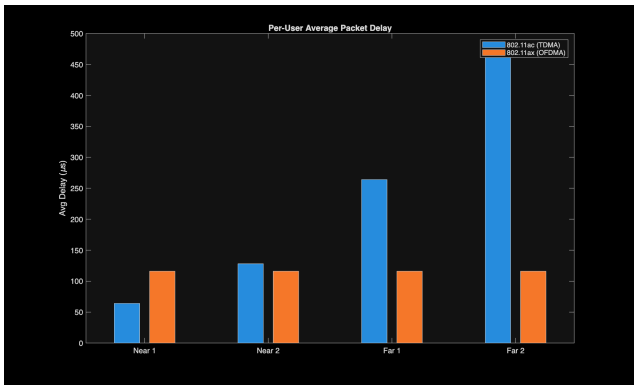

Figure 12: Scenario B: Per-user delay. 802.11ax imposes a constant ≈116 μs on all users. In 802.11ac, Far 2 (position 4 in round-robin) suffers ≈455 μs because it waits for three faster users in each TDMA round.

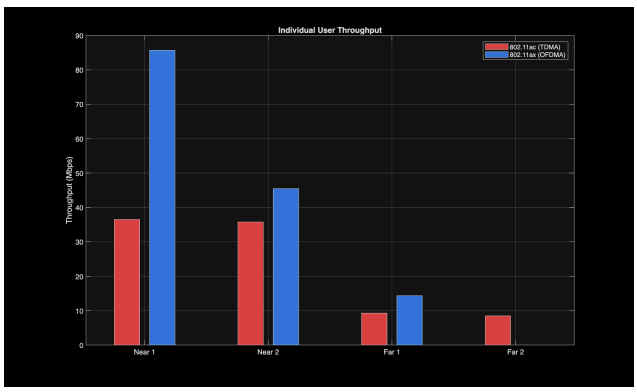

Figure 10: Scenario B: Per-user throughput. Near 1 reaches 85.8 Mbps in 802.11ax (MCS 11, 38 dB SNR, dedicated RU). Far 2 generates near-zero throughput in 802.11ax: MCS 2 at 8 dB in a fixed OFDMA slot dominated by the slowest RU. In 802.11ac both Near (≈36 Mbps) and Far (≈9 Mbps) users are more balanced.

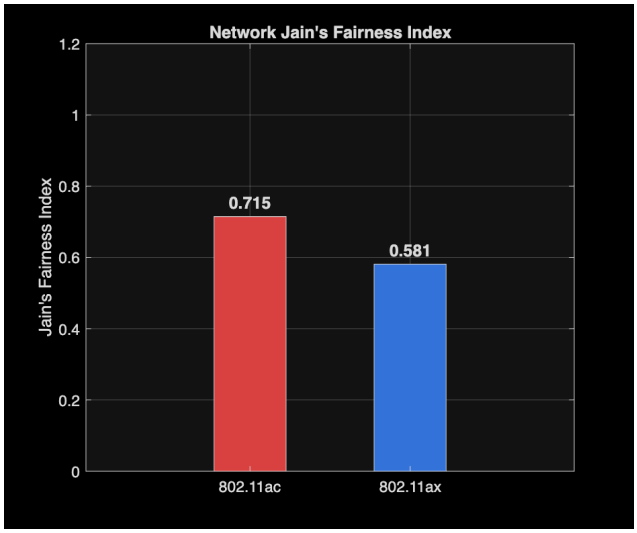

Figure 13: Scenario B: Jain's Fairness Index. Despite higher total capacity, 802.11ax (0.581) scores lower than 802.11ac (0.715). The static RU assignment leaves Far 2 nearly starved while Near 1 dominates.

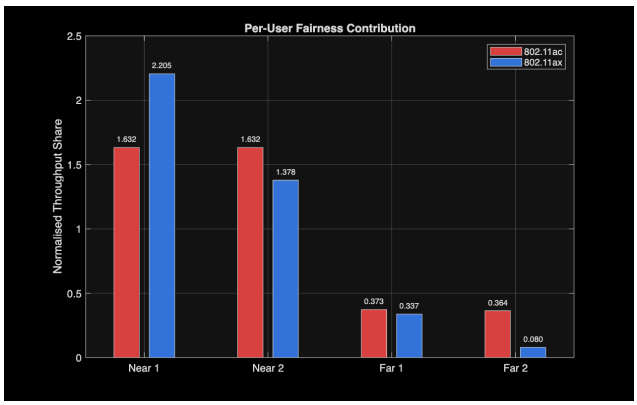

Figure 14: Scenario B: Per-user normalised throughput share. Near 1 in 802.11ax reaches 2.205, while Far 2 drops to 0.080 — a 27× ratio that explains the low Jain index. In 802.11ac the range is 0.364 to 1.632.

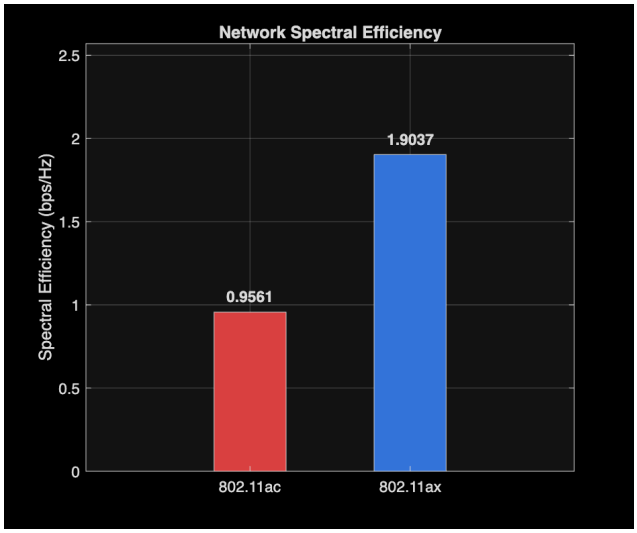

Figure 15: Scenario B: Network SE. 802.11ax achieves 1.904 bps/Hz versus 0.956 bps/Hz for 802.11ac, doubling spectral efficiency when Near users exploit their high-MCS RUs simultaneously.

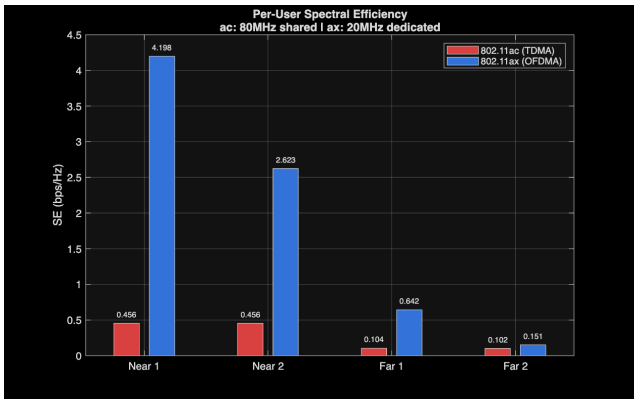

Figure 16: Scenario B: Per-user SE (ax referenced to 20 MHz sub-band; ac to 80 MHz shared). Near 1 in 802.11ax reaches ≈4.2 bps/Hz. Far 2 in 802.11ax drops to ≈0.15 bps/Hz, barely above the 802.11ac value.

4 Limitations

HE-SU workaround. The simulation uses `wlanHESUConfig` (80 MHz, SISO) per user instead of a true multi-user `wlanHEMUConfig` because MATLAB's WLAN Toolbox does not support `NumSpaceTimeStreams=1` in multi-RU SISO configurations. The OFDMA simultaneity is captured via time accounting (`ofdma_slot = max(slot_dur_ax)`), which is physically correct but approximates the sub-band PHY rather than simulating true 20 MHz RU waveforms. Each user's bit rate is computed at 80 MHz HE-SU capacity rather than 20 MHz RU capacity.

Simplified TDMA delay model. The queuing delay for user u is computed as $(u-1) \times T_u$, where T_u is the user's *own* slot duration. This assumes all preceding users have identical slot durations, which is incorrect when users differ in payload size (as in Scenario A). The actual wait depends on the sum of preceding users' slot durations. This causes the delay to be slightly underestimated for small-packet users that follow large-packet users.

Static RU allocation. RUs are not adapted between packets. Far 2 in Scenario B wastes its OFDMA slot every round due to very low MCS at 8 dB SNR. Real 802.11ax deployments use dynamic RU scheduling.

No link adaptation or retransmissions. MCS indices are fixed; packet errors are counted as drops without retransmission. This understates true delay and overstates throughput for low-SNR users.

Per-user SE reference bandwidth. The 802.11ax per-user SE (Figs. 8, 16) uses 20 MHz as the denominator while the throughput numerator reflects 80 MHz HE-SU simulation. This inflates per-user SE relative to a true 20 MHz RU and should be interpreted as a normalised rather than absolute measure.

5 Conclusions

Neither standard is universally superior. In **Scenario A** (heterogeneous traffic), 802.11ac doubles total throughput (108.8 vs. 54.4 Mbps) and network SE (1.360 vs. 0.756 bps/Hz), but 802.11ax reduces average delay by 44% (83 vs. 148 μs) and achieves higher Jain fairness (0.773 vs. 0.675). In **Scenario B** (variable SNR), 802.11ax nearly doubles capacity (145.5 vs. 77.4 Mbps) and eliminates extreme per-user delays (116 vs. up to 455 μs), but its static RU assignment starves Far 2, lowering the Jain index to 0.581 versus 0.715 for 802.11ac. These results confirm that OFDMA requires adaptive per-RU scheduling to reconcile its capacity gains with per-user fairness. The HE-SU workaround preserves the OFDMA time semantics and does not alter these qualitative conclusions.

6 References

References

[1] E. Khorov, A. Kiryanov, A. Lyakhov, and G. Bianchi, "A Tutorial on IEEE 802.11ax High Efficiency WLANs," *IEEE Commun. Surveys & Tutorials*, vol. 21, no. 1, pp. 197–216, 2019.

[2] B. Bellalta, "IEEE 802.11ax: High-efficiency WLANs," *IEEE Wireless Commun.*, vol. 23, no. 1, pp. 38–46, 2016.

[3] R. Jain, D. Chiu, and W. Hawe, "A Quantitative Measure of Fairness and Discrimination," *DEC Research Report TR-301*, 1984.

[4] MathWorks, "WLAN Toolbox — `wlanHESUConfig`, `wlanVHTConfig`," *MATLAB R2023b*, The MathWorks, Inc., 2023.